

Student Grade Management System

Project Title: Student Grade Management System

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GitHub Link: <https://github.com/aziz51151868-oss/student-grade-management-system-abdulaziz.git>

Program Logic

The program runs on a continuous while loop that keeps the menu open until the user types '6' to exit. It uses a Scanner to read input and a switch statement to run the chosen option. Since we couldn't use databases, the data is stored in parallel 1D arrays: one for strings (studentNames) and one for integers (studentGrades). A tracking variable called studentCount makes sure names and grades are always saved at the exact same index.

Major Functions Used

To make the code clean, I moved the logic out of the main method and into specific functions:

- displayMenu(): Prints the menu options.
- addStudent(): Checks if there is space in the array, validates the grade is between 0-100, and saves the data.
- displayAllStudents(): Loops through the arrays to print everything saved so far.
- searchStudent(): Uses a loop and .equalsIgnoreCase() to find a name and print the matching grade.
- calculateAverage(): Adds all the grades together and divides by the total number of students.
- findHighestAndLowest(): Loops through the grades array to track the index of the best and worst scores.

Screenshot of Sample Runs

Output

Welcome to Student Grade Management System

1. Add Student and Grade
2. Display All Students and Grades
3. Find a Student's Grade
4. Calculate Class Average
5. Find Highest and Lowest Grades
6. Exit

Enter your choice: 1

Enter student name: Alex

Enter student grade: 19

Student added successfully!

Welcome to Student Grade Management System

1. Add Student and Grade
2. Display All Students and Grades
3. Find a Student's Grade
4. Calculate Class Average
5. Find Highest and Lowest Grades
6. Exit

Enter your choice: 1

Enter student name: Bond

Enter student grade: 98

Student added successfully!

Welcome to Student Grade Management System

1. Add Student and Grade
2. Display All Students and Grades
3. Find a Student's Grade
4. Calculate Class Average
5. Find Highest and Lowest Grades
6. Exit

Enter your choice: 1

Enter student name: Alisa

Enter student grade: 76

Student added successfully!

Output

Welcome to Student Grade Management System

1. Add Student and Grade
2. Display All Students and Grades
3. Find a Student's Grade
4. Calculate Class Average
5. Find Highest and Lowest Grades
6. Exit

Enter your choice: 2

List of Students and Grades:

Alex - 19

Bond - 98

Alisa - 76

Welcome to Student Grade Management System

1. Add Student and Grade
2. Display All Students and Grades
3. Find a Student's Grade
4. Calculate Class Average
5. Find Highest and Lowest Grades
6. Exit

Enter your choice: 3

Enter student name: Alisa

Alisa's Grade: 76

Welcome to Student Grade Management System

1. Add Student and Grade
2. Display All Students and Grades
3. Find a Student's Grade
4. Calculate Class Average
5. Find Highest and Lowest Grades
6. Exit

Enter your choice: 4

Class Average: 64.33333333333333

```
Welcome to Student Grade Management System
```

1. Add Student and Grade
2. Display All Students and Grades
3. Find a Student's Grade
4. Calculate Class Average
5. Find Highest and Lowest Grades
6. Exit

```
Enter your choice: 5
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```
Highest Grade: Bond - 98
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Lowest Grade: Alex - 19
```

```
Welcome to Student Grade Management System
```

1. Add Student and Grade
2. Display All Students and Grades
3. Find a Student's Grade
4. Calculate Class Average
5. Find Highest and Lowest Grades
6. Exit

```
Enter your choice: 6
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Exiting system. Goodbye!
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=== Code Execution Successful ===
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Individual Contributions

I completed this project by myself. My contributions are:

- Writing the Java code and implementing the parallel 1D arrays logic.
- Creating the menu loop and refactoring the code into clean methods.
- Adding input validation to prevent crashes.
- Setting up the GitHub repository and tracking changes with Git commits.